Tues 8-10 am , $\boxed{\text{12pm - 2pm}}$

AST ① $b_n \geq 0$
② $\{b_n\}$ is decreasing on $n \geq N$ (positive integer)
③ $b_n \to 0$

Tut 5

Ex. 9.6

$(-1)^n b_n$

Q44 $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}+\sqrt{n+1}}$

$$\frac{1}{\sqrt{n}+\sqrt{n+1}} \sim \frac{1}{\sqrt{n+1}}$$

Let $a_n = \dfrac{(-1)^n}{\sqrt{n}+\sqrt{n+1}}$ for $n \geq 1$.

$\boxed{\displaystyle\sum_{n=1}^{\infty} |a_n|}$ div. by LCT with $c_n = \dfrac{1}{\sqrt{n+1}}$.

$\therefore \displaystyle\sum_{n=1}^{\infty} a_n$ conv. by AST.

$\therefore \displaystyle\sum_{n=1}^{\infty} a_n$ conv. cond.

2nd approach to show $\sum |a_n|$ div.

(Telescoping Series)

$$\sum |a_n| = \sum \frac{1}{\sqrt{n}+\sqrt{n+1}}$$

$$= \sum \frac{\sqrt{n}-\sqrt{n+1}}{n-(n+1)}$$

$$= \sum \sqrt{n+1}-\sqrt{n}$$

Q48 $\quad 1+\frac{1}{4}-\frac{1}{9}-\frac{1}{16}+\frac{1}{25}+\frac{1}{36}-\frac{1}{49}-\frac{1}{64}+\cdots$

$\cancel{\#}$
$\sum(-1)^n a_n$

Hard to describe using $\sum a_n$

$\#\ \sum |a_n|$ Converges $\Rightarrow \sum a_n$ converges

Let $1+\frac{1}{4}-\frac{1}{9}-\frac{1}{16}+\cdots=\sum_{n=1}^{\infty} a_n$ where
$$\left.\begin{aligned} a_1 &= 1 \\ a_2 &= \tfrac{1}{4} \\ a_3 &= -\tfrac{1}{9} \\ &\vdots \end{aligned}\right\} \text{A bit Informal}$$

More Formal:

$I_1=\{1,2,5,6,9,10,\ldots\}$

$I_2=\{3,4,7,8,11,12,\ldots\}$

$$a_n=\begin{cases} \frac{1}{n^2} & \text{if } n\in I_1 \\ -\frac{1}{n^2} & \text{if } n\in I_2 \end{cases}$$

Note that $|a_n|=\frac{1}{n^2}$ for $n\geq 1$.

$$\sum_{n=1}^{\infty}|a_n|=\sum_{n=1}^{\infty}\frac{1}{n^2}$$

Since $\sum_{n=1}^{\infty}\frac{1}{n^2}$ conv., $\sum_{n=1}^{\infty}|a_n|$ conv.

So $\sum_{n=1}^{\infty} a_n$ conv. abs.

Tut 6   Ex. 9.8E

"Building blocks": $\frac{x^n}{n!}$

Q31 $e^x \sin x = \ ?$ → First four nonzero terms.

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\sin x \overset{0+}{=} x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

	$\sin x$ \ e^x	1	1	$\frac{1}{2!}$	$\frac{1}{3!}$	$\frac{1}{4!}$	$\frac{1}{5!}$
x^0	0	0	0	0	0	0	0
x^1	1	1	1	$\frac{1}{2!}$	$\frac{1}{3!}$	$\frac{1}{4!}$	
x^2	0	0	0	0	0	0	
x^3	$-\frac{1}{3!}$	$-\frac{1}{3!}$	$-\frac{1}{3!}$	$-\frac{1}{2!3!}$	$\frac{1}{3!3!}$	$\frac{1}{3!4!}$	
x^4	0	0	0	0	0	0	
x^5	$\frac{1}{5!}$	$\frac{1}{5!}$	$\frac{1}{5!}$	$\cdots$			
	0						
	$-\frac{1}{7!}$	$\cdots$					

Const. x x^2 x^3

1   2   3   4

$$e^x \sin x = \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots\right)\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right)$$

$$= 0 + \underline{(1+0)x} + \underline{(0+1+0)x^2} + \underline{\left(\frac{1}{2!} - \frac{1}{3!}\right)x^3} + \left(\frac{1}{3!} - \frac{1}{3!}\right)x^4 + \underline{\left(\frac{1}{4!} - \frac{1}{2!3!} + \frac{1}{5!}\right)}x^5 + \cdots$$

<u>Q33</u> $(\tan^{-1} x)^2$

$$\tan^{-1} x = x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \frac{1}{7}x^7 + \cdots$$